2.1 What Is Diffusion?

Biology · Movement of Substances · Syllabus refs (G2 & G3): 2a

Name: _____ Class: _____ Date: _____

Learning outcomes

- Define diffusion and describe its role in nutrient uptake and gaseous exchange in plants and humans.
- Describe the factors that affect the rate of diffusion.

Trigger activity

Watch the animation. Someone comes out of the toilet at one end of a corridor. A little later, a person waiting at the other end can smell it. **How did the smell get there, when nobody carried it?**

The same thing happens with perfume. Someone sprays perfume at one end of a room, and a few moments later a person at the other end can smell it.

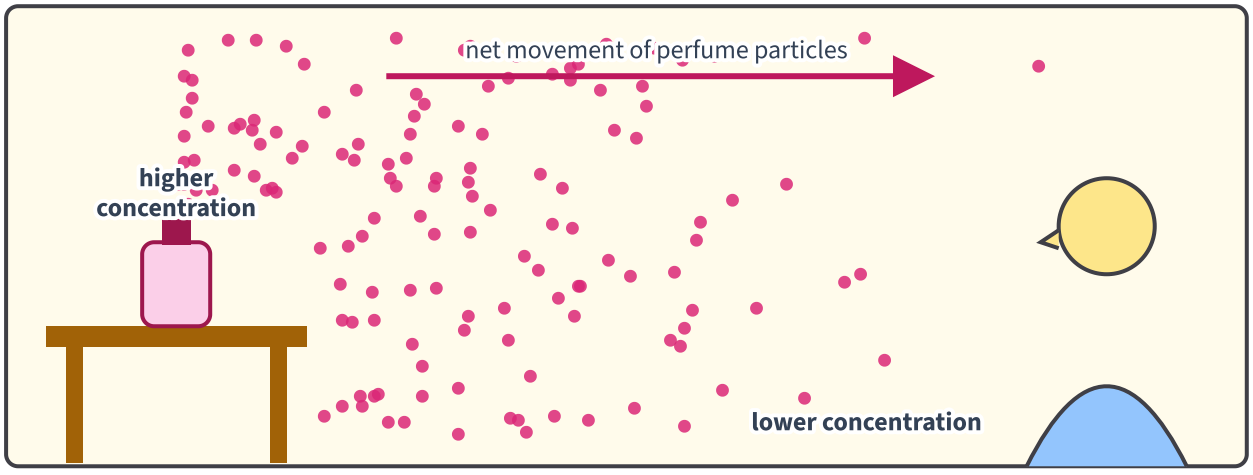


Fig. 2.1 — Perfume particles spread out from where they are crowded to where there are fewer of them.

The perfume particles are always moving about at random, bumping into air particles. Near the bottle there are many perfume particles; far away there are few. So, overall, more particles move *away* from the bottle than towards it. The perfume spreads until it is evenly mixed in the air.

What is diffusion?

The difference in concentration between two regions is called a **concentration gradient**. Particles spread from the region of higher concentration to the region of lower concentration: we say they move **down the concentration gradient**.

Diffusion is the net movement of particles (atoms, molecules or ions) from a region where they are at a higher concentration to a region where they are at a lower concentration, that is, down a concentration gradient.

> **Key idea:** the **steeper** the concentration gradient (the bigger the difference in concentration), the **faster** the rate of diffusion.

Why "net"? Each particle moves at random, in any direction. Some particles even move back towards the crowded region. The *net* movement is the overall result: more particles go one way than the other.

Practical: diffusion of a dissolved substance in water

Aim: to investigate how a dissolved substance spreads through water.

Procedure

1. Gently drop one crystal of **copper(II) sulfate** into a gas jar filled with water. Cover the jar.
2. Leave the jar to stand, without shaking it, for a few days.
3. Observe how the colour of the water changes.

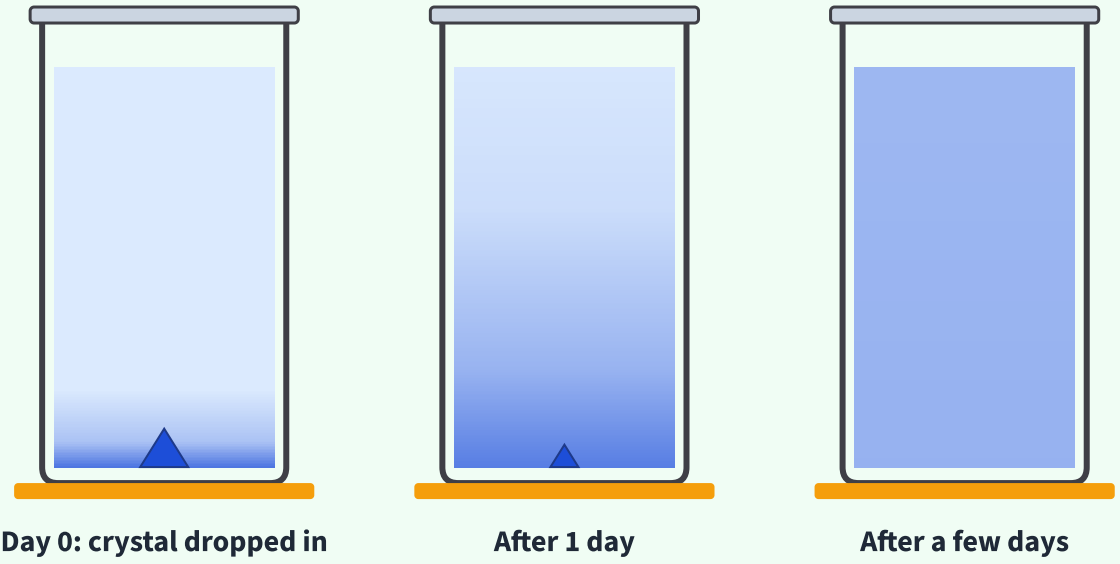


Fig. 2.2 — The copper(II) sulfate crystal dissolves, and the blue colour slowly spreads through all the water.

Observation and discussion

1. Use the idea of a concentration gradient to explain why the blue colour slowly spreads through the water.
2. When does diffusion stop?

In the practical, the copper(II) sulfate particles moved from the region of higher concentration around the crystal to regions of lower concentration in the water. The solute diffused until it was evenly spread, forming a solution.

If **more than one substance** is dissolved in the same liquid, each one diffuses on its own, down its own concentration gradient. The particles of one substance do not affect the diffusion of the other.

Diffusion across a membrane

A **permeable membrane** lets both the solvent (water) and the solutes (the dissolved substances) pass through it. In Fig. 2.3, a permeable membrane divides a beaker into two halves:

- the **left** side has a higher concentration of **potassium nitrate** (yellow);
- the **right** side has a higher concentration of **copper(II) sulfate** (blue).

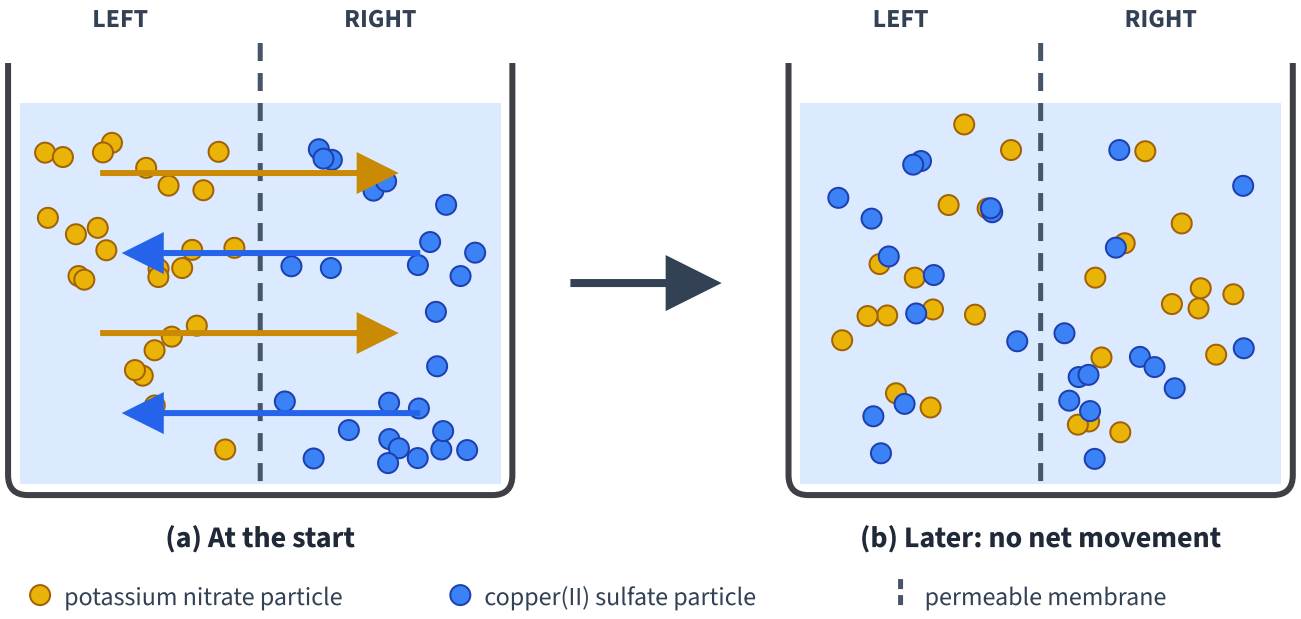


Fig. 2.3 — Potassium nitrate and copper(II) sulfate particles diffusing through a permeable membrane.

So the potassium nitrate particles diffuse across the membrane to the right, and the copper(II) sulfate particles diffuse across to the left. In the end, the concentration of each substance is the **same on both sides** of the membrane.

Quick check

Q1. Which statement describes diffusion?

- ○ Net movement of particles from a region of lower concentration to a region of higher concentration
- ○ Net movement of particles from a region of higher concentration to a region of lower concentration
- ○ Movement of water only, through a partially permeable membrane
- ○ Movement of particles only through a membrane

Q2. Region X has a high concentration of a substance and region Y has a low concentration. In which direction is the net movement of the substance?

- ○ From Y to X
- ○ From X to Y
- ○ There is no movement
- ○ Equally in both directions

Q3. A crystal of copper(II) sulfate is left in a gas jar of water. After many days the water is evenly blue. Which is correct?

- ○ The copper(II) sulfate particles have stopped moving.
- ○ There is no net movement, but the particles are still moving.
- ○ The particles are now moving towards the bottom of the jar.
- ○ The water has turned into copper(II) sulfate.

Q4. A beaker is divided by a permeable membrane. The left side has more potassium nitrate; the right side has more copper(II) sulfate. What happens?

- ○ Only potassium nitrate moves.
- ○ Potassium nitrate moves to the right and copper(II) sulfate moves to the left.
- ○ Both substances move to the left.
- ○ Nothing moves, because they block each other.

Factors that affect the rate of diffusion

How fast a substance diffuses depends on three factors:

1. concentration gradient | **2. diffusion distance** | **3. surface area to volume ratio**

1. Concentration gradient

The **steeper** the concentration gradient (the bigger the difference in concentration between the two regions), the **faster** the rate of diffusion. When there is no concentration gradient, there is no net diffusion.

2. Diffusion distance

The distance a substance has to travel is the **diffusion distance**. The **shorter** the diffusion distance, the less time the substance needs to get there, so the **faster** the rate of diffusion.

3. Surface area to volume ratio

Substances enter and leave a cell through its surface (the cell membrane), so the rate of diffusion depends on the cell's **surface area**. For two cells with the same volume, the one with the larger surface area takes substances in faster. Comparing a cell's surface area with its volume gives its **surface area to volume ratio** (SA : V).

> The **greater** the surface area to volume ratio, the **faster** the rate of diffusion into and out of the cell.

As a cell gets bigger, its volume grows faster than its surface area, so its SA : V gets smaller. This is why cells are usually small, and why a large organism is made of many small cells rather than one giant cell.

Try it again: the shape builder and the agar-cube experiment on SA : V are in Lesson 1.2.

A. Cube X has sides of 1 cm: surface area 6 cm^2, volume 1 cm^3. Cube Y has sides of 3 cm: surface area 54 cm^2, volume 27 cm^3. Work out the SA : V of cube Y (as a number : 1).

Your answer (: 1)

Diffusion in living things

The cell membrane is **partially permeable**: it lets some substances through but not others. Small molecules such as **oxygen**, **carbon dioxide**, **glucose** and **amino acids** can pass through;

large molecules such as proteins cannot. Living things use diffusion to take in what they need and get rid of what they do not. Notice how each example has a **short diffusion distance** or a **large surface area**.

In humans

1. Gas exchange in the lungs. The air in the lungs has a high concentration of oxygen, and the blood arriving there has a low concentration. Oxygen diffuses from the air space, through a very thin wall, into the red blood cells. Carbon dioxide diffuses the other way, from the blood into the air space, and is breathed out. The wall is only **one cell thick**, so the diffusion distance is short and diffusion is fast.

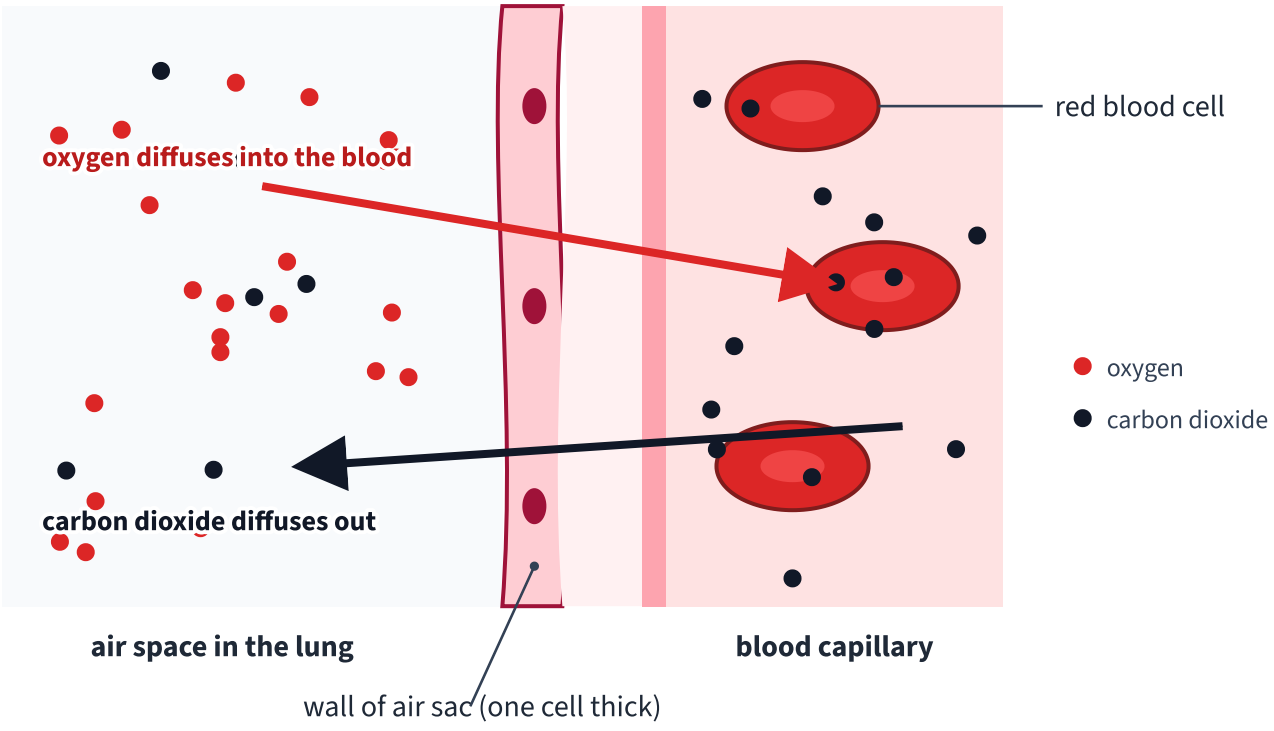
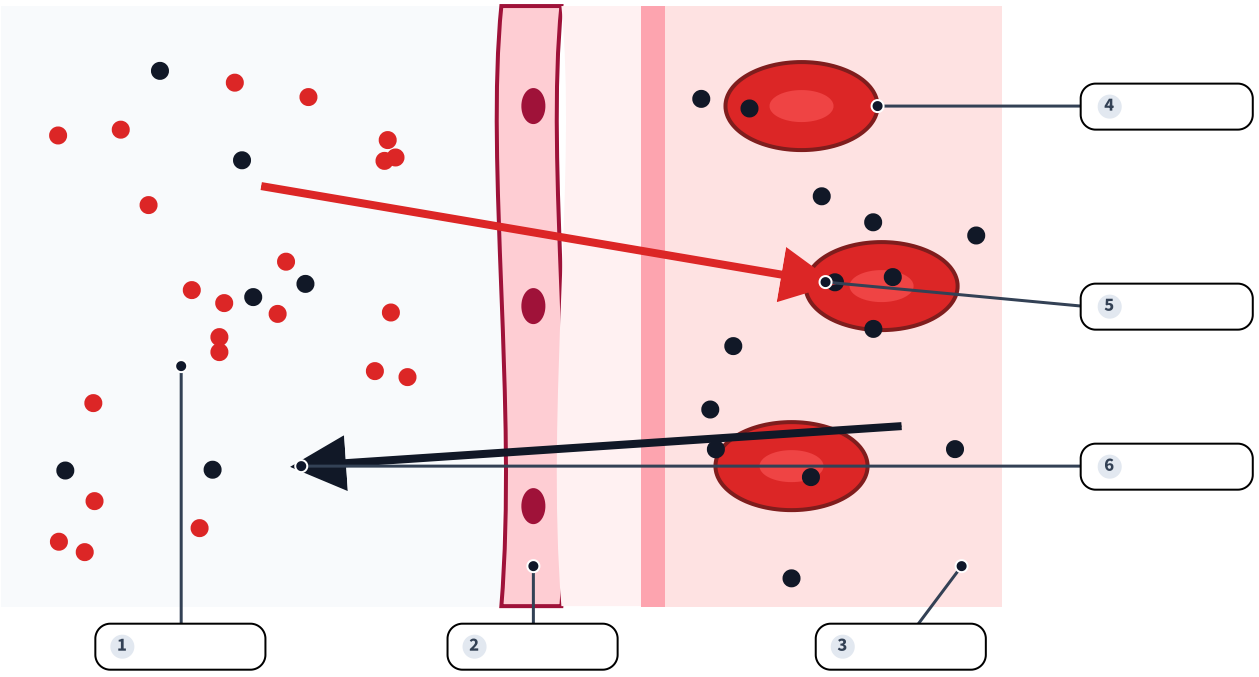


Fig. 2.4 — Gas exchange between an air space in the lung and the blood.

Label gas exchange in the lungs.

chloroplast | wall of air sac | oxygen | red blood cell | blood capillary | air space in the lung | cell wall | carbon dioxide



Gas exchange in the lungs.

2. Absorption of digested food in the small intestine. After digestion, there is a high concentration of small food molecules, such as **glucose** and **amino acids**, inside the small intestine, and a lower concentration in the blood. They diffuse through the lining of the small intestine into the blood capillaries, and the blood carries them around the body. The lining is one cell thick (short diffusion distance) and has many tiny folds (large surface area).

> Glucose and amino acids are **solutes**: substances dissolved in another substance (the **solvent**). Together they make a **solution**.

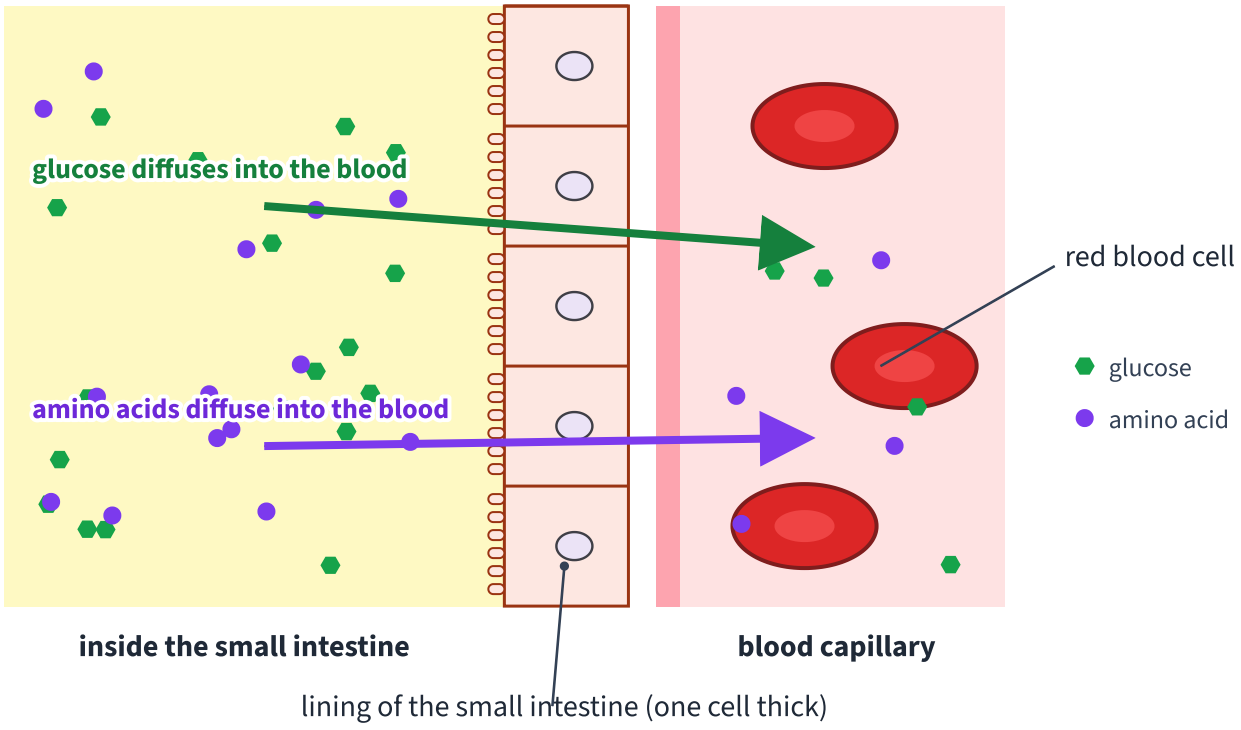


Fig. 2.5 — Absorption of digested food by diffusion in the small intestine.

In plants

1. Gas exchange through the stomata. Leaves have tiny pores called **stomata** (one is a **stoma**), mostly on the underside. In the light, leaf cells use carbon dioxide for **photosynthesis**, so the concentration of carbon dioxide inside the leaf is low. **Carbon dioxide** diffuses from the air **into** the leaf through the stomata. Photosynthesis makes **oxygen**, so its concentration inside the leaf is high, and oxygen diffuses **out** of the leaf through the stomata.

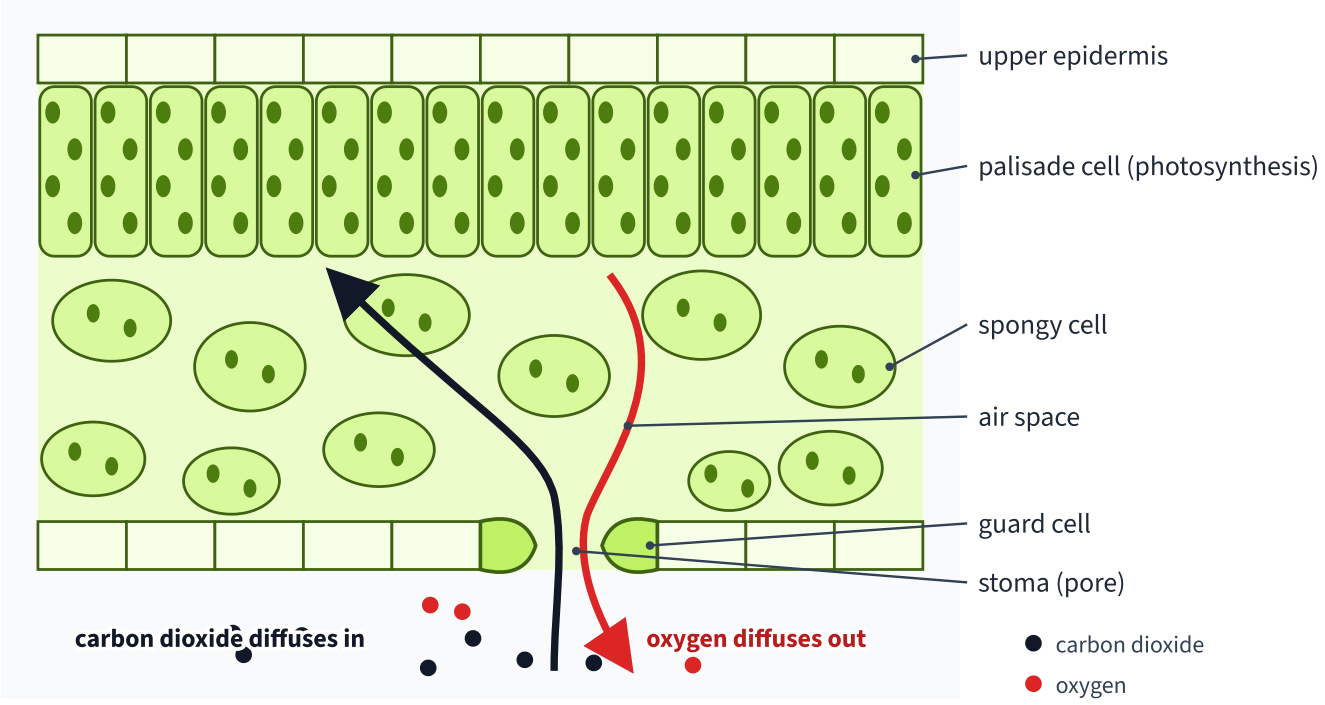
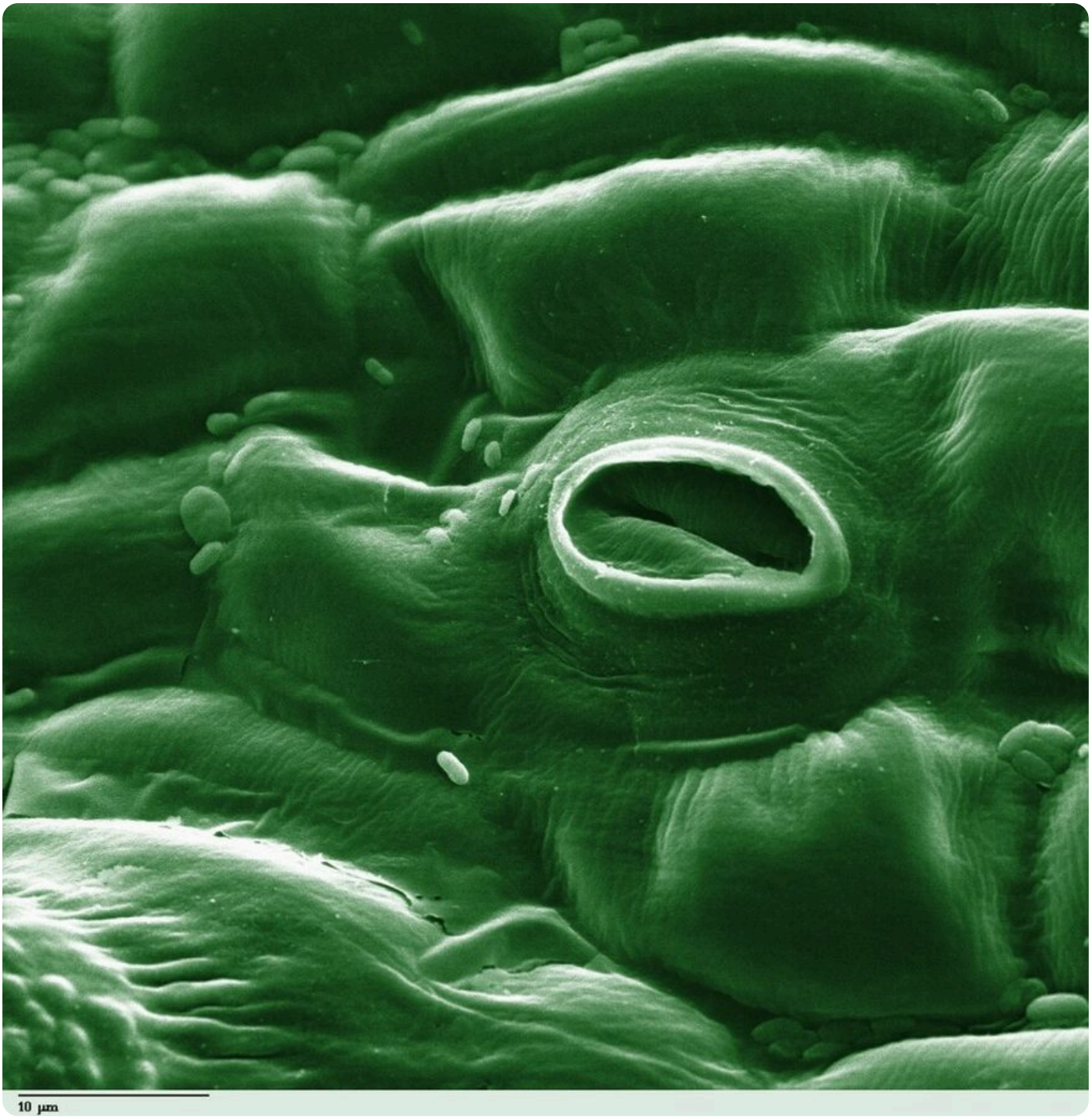


Fig. 2.6 — Carbon dioxide diffuses into a leaf and oxygen diffuses out through a stoma.



A real stoma on a tomato leaf (scanning electron micrograph, colour added). The pore is between two guard cells.

Image: Photohound, public domain, via Wikimedia Commons.

2. Root hair cell. A root hair cell has a **long, narrow protrusion** (the root hair) that grows out between the soil particles. This gives the cell a **large surface area to volume ratio**, so substances diffuse into it faster. Oxygen diffuses in from the air spaces in the soil (and carbon dioxide diffuses out). Mineral salts dissolved in the soil water can also diffuse in when their concentration in the soil is higher than in the cell. (Water enters by osmosis: see Lesson 2.2.)

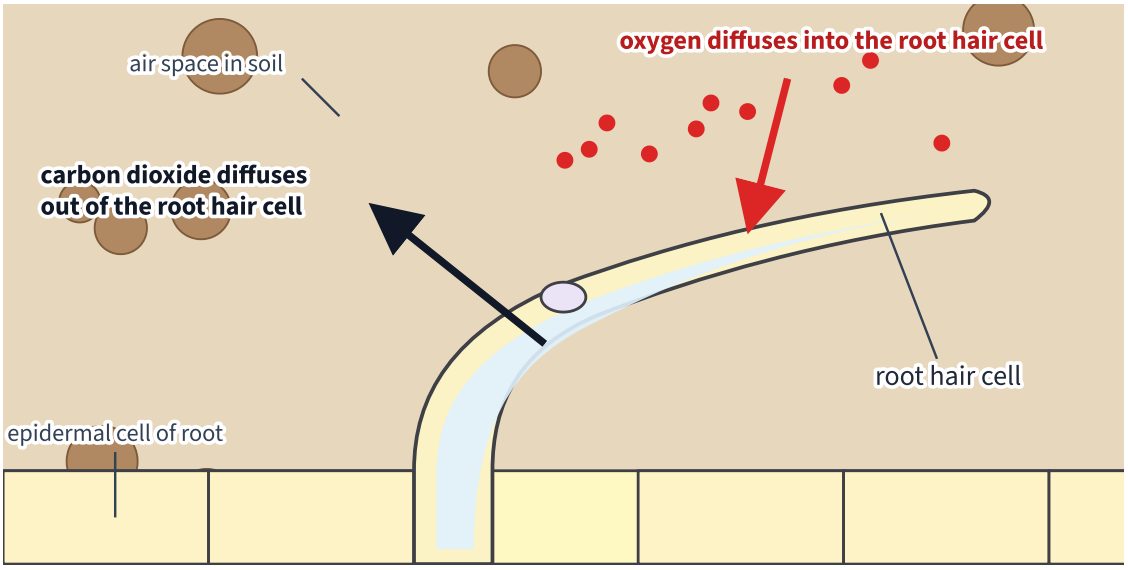


Fig. 2.7 — A root hair cell: its long, narrow protrusion gives a large surface area for diffusion.

Big idea: systems

Living things depend on diffusion to exchange gases and take in nutrients. You will meet these examples again in the topics on nutrition, transport and respiration.

Check your understanding

1. Which change makes diffusion faster?

- ○ A shallower concentration gradient
- ○ A longer diffusion distance
- ○ A smaller surface area to volume ratio
- ○ A steeper concentration gradient

2. Why are most cells very small?

- ○ Small cells have a smaller surface area to volume ratio.
- ○ Small cells have a larger surface area to volume ratio, so substances can move in and out fast enough.
- ○ Small cells do not need oxygen.
- ○ Small cells have no cell membrane.

3. In the lungs, in which direction does oxygen diffuse?

- ○ From the blood into the air space
- ○ From the air space into the blood
- ○ It does not move
- ○ From the red blood cells into the air space

4. How do glucose and amino acids get from the small intestine into the blood?

- ○ By diffusion into the blood capillaries
- ○ By photosynthesis
- ○ By breathing
- ○ They stay in the small intestine

5. In the light, which gases diffuse into and out of a leaf through the stomata?

- ○ Oxygen in, carbon dioxide out
- ○ Carbon dioxide in, oxygen out
- ○ Nitrogen in, oxygen out
- ○ Water in, glucose out

6. How does the long, narrow protrusion of a root hair cell help it?

- ○ It gives the cell a fixed shape.
- ○ It gives a large surface area to volume ratio, so substances diffuse in faster.
- ○ It makes food by photosynthesis.
- ○ It stops water entering the cell.

7. Which substance cannot pass through a cell membrane?

- ○ Oxygen
- ○ Carbon dioxide
- ○ Glucose
- ○ Protein